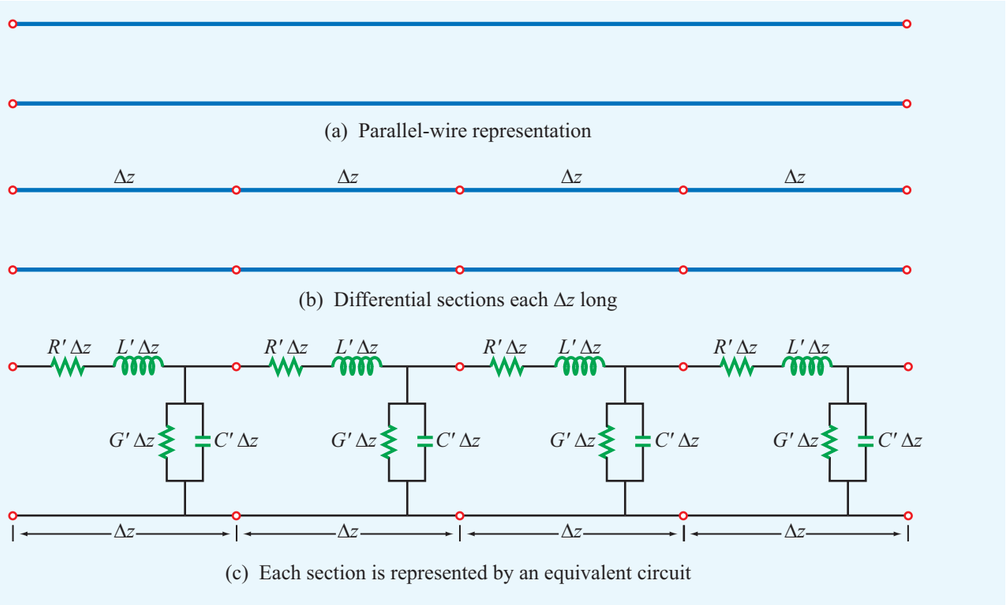


Figure 2-6 Regardless of its cross-sectional shape, a TEM transmission line is represented by the parallel-wire configuration shown in part (a). To obtain equations relating voltages and currents, the line is subdivided into small differential sections in part (b), each of which is then represented by an equivalent circuit in part (c).

each of these lines, the expressions are functions of two sets of parameters: (1) geometric parameters defining the cross-sectional dimensions of the given line and (2) the electromagnetic constitutive parameters of the conducting and insulating materials. The pertinent geometric parameters are:

• *Coaxial line* (**Fig. 2-4(a)**):

a = outer radius of inner conductor, m
b = inner radius of outer conductor, m

• *Two-wire line* (**Fig. 2-4(b)**):

d = diameter of each wire, m
D = spacing between wires' centers, m

• *Parallel-plate line* (**Fig. 2-4(c)**):

w = width of each plate, m
h = thickness of insulation between plates, m

► The pertinent ***constitutive parameters*** apply to all three lines and consist of two groups:

(1) μ_c and σ_c are the magnetic permeability and electrical conductivity of the conductors, and

(2) ε, μ, and σ are the electrical permittivity, magnetic permeability, and electrical conductivity of the insulation material separating them. ◄

Appendix B contains tabulated values for these constitutive parameters for various materials. For the purposes of this chapter, we need not concern ourselves with the derivations leading to the expressions in **Table 2-1**. The techniques necessary for computing R', L', G', and C' for the general case of an arbitrary two-conductor configuration are presented in later chapters.

The lumped-element model shown in **Fig. 2-6(c)** reflects the physical phenomena associated with the currents and voltages on any TEM transmission line. It consists of two in-series elements, R' and L', and two shunt elements, G' and C'. To

2-2 LUMPED-ELEMENT MODEL 51

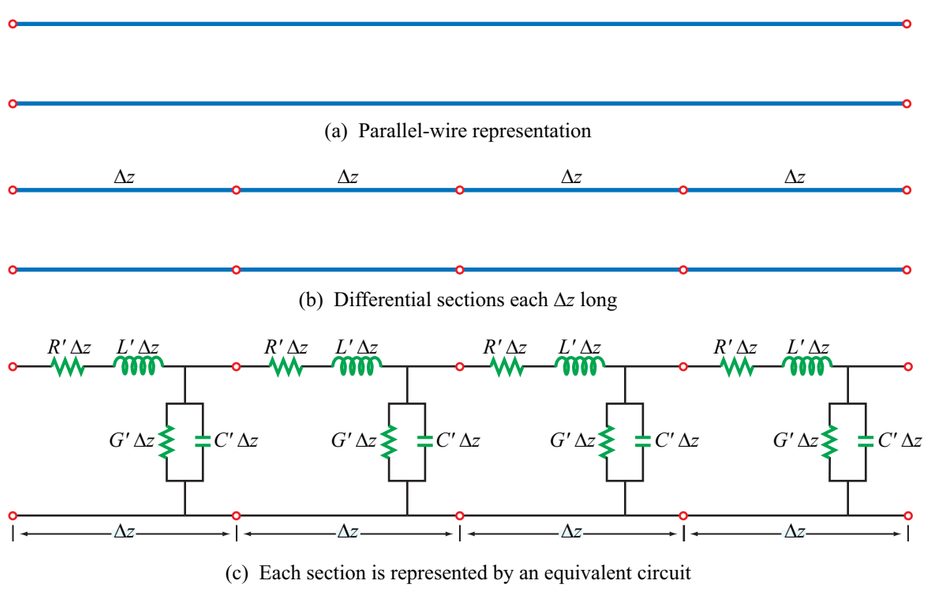


图 2-6 无论其横截面形状如何，TEM 传输线都由 (a) 部分所示的平行线配置表示。为了获得与电压和电流相关的方程，在（b）部分中将线路细分为小的微分部分，然后每个部分由（c）部分中的等效电路表示。

对于每条线，表达式都是两组参数的函数：（1）定义给定线的横截面尺寸的几何参数和（2）导电和绝缘材料的电磁本构参数。相关的几何参数是：

• 同轴线（图2-4（a））：

a = 内导体外半径，m b = 外导体内半径，m

• 两线制线路（图2-4（b））：

d = 每根导线的直径，m D = 导线中心之间的间距，m

• 平行板线（图2-4（c））：

w = 每块板的宽度，m h = 板之间绝缘层的厚度，m

► 相关的本构参数适用于所有三条线，并由两组组成：

(1) μ_c 和 σ_c 是导体的磁导率和电导率，并且

(2) ε、μ 和 σ 是分隔它们的绝缘材料的介电常数、磁导率和电导率。 ◄

附录 B 包含各种材料的这些本构参数的表格值。为了本章的目的，我们不需要关心表 2-1 中表达式的推导。对于任意两导体配置的一般情况，计算 R′、L′、G′ 和 C′ 所需的技术将在后面的章节中介绍。

图 2-6(c) 所示的集总元件模型反映了与任何 TEM 传输线上的电流和电压相关的物理现象。它由两个串联元件 R′ 和 L′ 以及两个并联元件 G′ 和 C′ 组成。到

Table 2-1 Transmission-line parameters R', L', G', and C' for three types of lines.

Parameter	Coaxial	Two-Wire	Parallel-Plate	Unit
R'	$\frac{R_s}{2\pi}\left(\frac{1}{a}+\frac{1}{b}\right)$	$\frac{2R_s}{\pi d}$	$\frac{2R_s}{w}$	Ω/m
L'	$\frac{\mu}{2\pi}\ln(b/a)$	$\frac{\mu}{\pi}\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right]$	$\frac{\mu h}{w}$	H/m
G'	$\frac{2\pi\sigma}{\ln(b/a)}$	$\frac{\pi\sigma}{\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right]}$	$\frac{\sigma w}{h}$	S/m
C'	$\frac{2\pi\varepsilon}{\ln(b/a)}$	$\frac{\pi\varepsilon}{\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right]}$	$\frac{\varepsilon w}{h}$	F/m

Notes: (1) Refer to Fig. 2-4 for definitions of dimensions. (2) μ, ε, and σ pertain to the insulating material between the conductors. (3) $R_s = \sqrt{\pi f \mu_c/\sigma_c}$. (4) μ_c and σ_c pertain to the conductors. (5) If $(D/d)^2 \gg 1$, then $\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right] \approx \ln(2D/d)$.

explain the lumped-element model, consider a small section of a coaxial line, as shown in Fig. 2-7. The line consists of inner and outer conductors of radii a and b separated by a material with permittivity ε, permeability μ, and conductivity σ. The two metal conductors are made of a material with conductivity σ_c and permeability μ_c.

Resistance R'

When a voltage source is connected across the terminals connected to the two conductors at the sending end of the line, currents flow through the conductors, primarily along the outer surface of the inner conductor and the inner surface of the outer conductor. The line resistance R' accounts for the combined resistance per unit length of the inner and outer conductors. The expression for R' is derived in Chapter 7 and is given by Eq. (7.96) as

$$R' = \frac{R_s}{2\pi}\left(\frac{1}{a}+\frac{1}{b}\right) \quad \textbf{(coax line)} \quad (\Omega/\text{m}), \qquad (2.5)$$

where R_s, which represents the ***surface resistance*** of the conductors, is given by Eq. (7.92a) as

$$R_s = \sqrt{\frac{\pi f \mu_c}{\sigma_c}} \quad (\Omega). \qquad (2.6)$$

The surface resistance depends not only on the material properties of the conductors (σ_c and μ_c), but also on the frequency f of the wave traveling on the line.

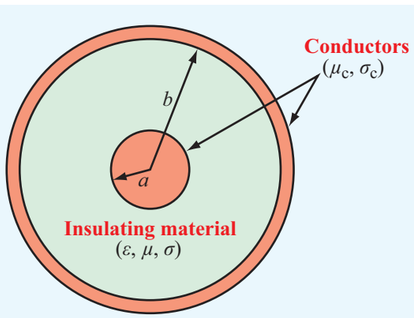


Figure 2-7 Cross section of a coaxial line with inner conductor of radius a and outer conductor of radius b. The conductors have magnetic permeability μ_c and conductivity σ_c, and the spacing material between the conductors has permittivity ε, permeability μ, and conductivity σ.

► For a ***perfect conductor*** with $\sigma_c = \infty$ or a high-conductivity material such that $(f\mu_c/\sigma_c) \ll 1$, R_s approaches zero, and so does R'. ◄

表 2-1 三种类型线路的传输线参数 R′、L′、G′ 和 C′。

Parameter	Coaxial	Two-Wire	Parallel-Plate	Unit
R'	$\frac{R_s}{2\pi}\left(\frac{1}{a}+\frac{1}{b}\right)$	$\frac{2R_s}{\pi d}$	$\frac{2R_s}{w}$	Ω/m
L'	$\frac{\mu}{2\pi}\ln(b/a)$	$\frac{\mu}{\pi}\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right]$	$\frac{\mu h}{w}$	H/m
G'	$\frac{2\pi\sigma}{\ln(b/a)}$	$\frac{\pi\sigma}{\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right]}$	$\frac{\sigma w}{h}$	S/m
C'	$\frac{2\pi\varepsilon}{\ln(b/a)}$	$\frac{\pi\varepsilon}{\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right]}$	$\frac{\varepsilon w}{h}$	F/m

Notes: (1) Refer to **Fig. 2-4** for definitions of dimensions. (2) μ, ε, and σ pertain to the insulating material between the conductors. (3) $R_s = \sqrt{\pi f \mu_c/\sigma_c}$. (4) μ_c and σ_c pertain to the conductors. (5) If $(D/d)^2 \gg 1$, then $\ln\left[(D/d)+\sqrt{(D/d)^2-1}\right] \approx \ln(2D/d)$.

为了解释集总元件模型，考虑同轴线的一小部分，如图 2-7 所示。该线路由半径为 a 和 b 的内导体和外导体组成，由具有介电常数 ε、磁导率 μ 和电导率 σ 的材料分隔开。两个金属导体由电导率σ_c和磁导率μ_c的材料制成。

电阻R′

当电压源跨接在线路发送端连接到两个导体的端子上时，电流主要沿着内导体的外表面和外导体的内表面流过导体。线路电阻R′是内导体和外导体每单位长度的组合电阻。 R′ 的表达式在第 7 章中推导，由式 (1) 给出。 (7.96) 作为

$$R' = \frac{R_s}{2\pi}\left(\frac{1}{a}+\frac{1}{b}\right) \quad \textbf{(coax line)} \quad (\Omega/\text{m}), \qquad (2.5)$$

其中 R_s 表示导体的表面电阻，由方程 1 给出。 (7.92a) 为

$$R_s = \sqrt{\frac{\pi f \mu_c}{\sigma_c}} \quad (\Omega). \qquad (2.6)$$

表面电阻不仅取决于导体的材料特性（σ_c 和 μ_c），还取决于在线路上传播的波的频率 f。

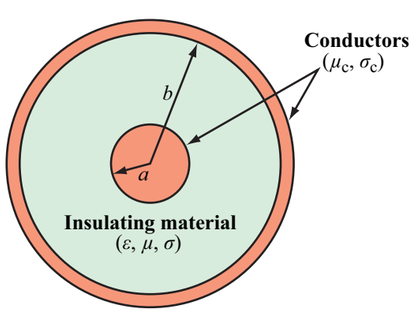


图 2-7 内导体半径为 a、外导体半径为 b 的同轴线横截面。导体具有磁导率μ_c和电导率σ_c，并且导体之间的间隔材料具有介电常数ε、磁导率μ 和电导率σ。

► 对于具有 $\sigma_c = \infty$ 的理想导体或 $(f\mu_c/\sigma_c) \ll 1$ 的高电导率材料，R_s 接近零，R′ 也接近零。 ◄

Inductance L'

Next, let us examine the line inductance L', which accounts for the joint inductance of both conductors. Application of Ampère's law in Chapter 5 to the definition of inductance leads to the following expression [Eq. (5.99)] for the inductance per unit length of a coaxial line:

$$L' = \frac{\mu}{2\pi}\ln\left(\frac{b}{a}\right) \quad \textbf{(coax line)} \quad \text{(H/m)}. \tag{2.7}$$

Conductance G'

The line conductance G' accounts for current flow between the outer and inner conductors, made possible by the conductivity σ of the insulator. It is precisely because the current flow is from one conductor to the other that G' appears as a *shunt* element in the lumped-element model. For the coaxial line, the conductance per unit length is given by Eq. (4.86) as

$$G' = \frac{2\pi\sigma}{\ln(b/a)} \quad \textbf{(coax line)} \quad \text{(S/m)}. \tag{2.8}$$

► If the material separating the inner and outer conductors is a ***perfect dielectric*** with $\sigma = 0$, then $G' = 0$. ◄

Capacitance C'

The last line parameter on our list is the line capacitance C'. When equal and opposite charges are placed on any two noncontacting conductors, a voltage difference develops between them. Capacitance is defined as the ratio of the charge to the voltage difference. For the coaxial line, the capacitance per unit length is given by Eq. (4.127) as

$$C' = \frac{2\pi\varepsilon}{\ln(b/a)} \quad \textbf{(coax line)} \quad \text{(F/m)}. \tag{2.9}$$

All TEM transmission lines share the following useful relations:

$$L'C' = \mu\varepsilon \quad \textbf{(all TEM lines)}, \tag{2.10}$$

and

$$\frac{G'}{C'} = \frac{\sigma}{\varepsilon} \quad \textbf{(all TEM lines)}. \tag{2.11}$$

If the insulating medium between the conductors is air, the transmission line is called an ***air line*** (e.g., coaxial air line or two-wire air line). For an air line, $\varepsilon = \varepsilon_0 = 8.854 \times 10^{-12}$ F/m, $\mu = \mu_0 = 4\pi \times 10^{-7}$ H/m, $\sigma = 0$, and $G' = 0$.

Concept Question 2-1: What is a transmission line? When should transmission-line effects be considered, and when may they be ignored?

Concept Question 2-2: What is the difference between dispersive and nondispersive transmission lines? What is the practical significance of dispersion?

Concept Question 2-3: What constitutes a TEM transmission line?

Concept Question 2-4: What purpose does the lumped-element circuit model serve? How are the line parameters R', L', G', and C' related to the physical and electromagnetic constitutive properties of the transmission line?

Exercise 2-1: Use **Table 2-1** to evaluate the line parameters of a two-wire air line with wires of radius 1 mm, separated by a distance of 2 cm. The wires may be treated as perfect conductors with $\sigma_c = \infty$.

Answer: $R' = 0$, $L' = 1.20$ (μH/m), $G' = 0$, $C' = 9.29$ (pF/m). (See EM.)

Exercise 2-2: Calculate the transmission line parameters at 1 MHz for a coaxial air line with inner and outer conductor diameters of 0.6 cm and 1.2 cm, respectively. The conductors are made of copper (see Appendix B for μ_c and σ_c of copper).

Answer: $R' = 2.07 \times 10^{-2}$ (Ω/m), $L' = 0.14$ (μH/m), $G' = 0$, $C' = 80.3$ (pF/m). (See EM.)

2-3 Transmission-Line Equations

A transmission line usually connects a source on one end to a load on the other. Before considering the complete circuit, however, we will develop general equations that describe the voltage across and current carried by the transmission line as a function of time t and spatial position z. Using the lumped-element model of **Fig. 2-6(c)**, we begin by considering a differential length Δz as shown in **Fig. 2-8**. The quantities $\upsilon(z,t)$ and $i(z,t)$ denote the instantaneous voltage and current at the left end of the differential section (node N), and similarly $\upsilon(z+\Delta z,t)$ and $i(z+\Delta z,t)$ denote the same quantities at node

电感L′

接下来，让我们检查线路电感 L′，它是两个导体的联合电感。应用第 5 章中的安培环路定律来定义电感，可得出以下表达式 [Eq.1]。 (5.99)] 同轴线单位长度的电感：

$$L' = \frac{\mu}{2\pi}\ln\left(\frac{b}{a}\right) \quad \textbf{(coax line)} \quad \text{(H/m)}. \tag{2.7}$$

电导G′

线路电导 G′ 说明了外导体和内导体之间的电流，这是由绝缘体的电导率 σ 实现的。正是因为电流从一个导体流向另一个导体，G′ 在集总元件模型中显示为分流元件。对于同轴线，单位长度的电导由式（1）给出。 (4.86) 作为

$$G' = \frac{2\pi\sigma}{\ln(b/a)} \quad \textbf{(coax line)} \quad \text{(S/m)}. \tag{2.8}$$

►如果分隔内导体和外导体的材料是理想电介质且 σ = 0，则 G′ = 0。 ◄

电容C′

我们列表中的最后一个线路参数是线路电容 C′。当相等且相反的电荷放置在任何两个非接触导体上时，它们之间就会产生电压差。电容定义为电荷与电压差之比。对于同轴线，单位长度的电容由式（1）给出。 (4.127) 作为

$$C' = \frac{2\pi\varepsilon}{\ln(b/a)} \quad \textbf{(coax line)} \quad \text{(F/m)}. \tag{2.9}$$

所有 TEM 传输线都具有以下有用关系：

$$L'C' = \mu\varepsilon \quad \textbf{(all TEM lines)}, \tag{2.10}$$

和

$$\frac{G'}{C'} = \frac{\sigma}{\varepsilon} \quad \textbf{(all TEM lines)}. \tag{2.11}$$

如果导体之间的绝缘介质是空气，则传输线称为空气线（例如同轴空气线或两线空气管路）。对于航空公司，$\varepsilon = \varepsilon_0 = 8.854 \times 10^{-12}$ F/m、$\mu = \mu_0 = 4\pi \times 10^{-7}$ H/m、$\sigma = 0$ 和 $G' = 0$。

概念问题2-1：什么是传输线？什么时候应该考虑传输线效应，什么时候可以忽略它们？

概念问题 2-2：色散传输线和非色散传输线有什么区别？色散的实际意义是什么？

概念问题 2-3：TEM 传输线由什么构成？

概念问题 2-4：集总元件电路模型有什么用途？线路参数 R′、L′、G′ 和 C′ 与传输线的物理和电磁本构特性有何关系？

练习 2-1：使用表 2-1 评估两线空气管路的线路参数，该两线空气管路的导线半径为 1 毫米，间隔距离为 2 厘米。电线可以被视为具有$\sigma_c = \infty$的理想导体。

答案：R′ = 0、L′ = 1.20 (μ H/m)、G′ = 0、C′ = 9.29 (pF/m)。 （参见EM。）

练习 2-2：计算内导体直径和外导体直径分别为 0.6 厘米和 1.2 厘米的同轴空气线路在 1 MHz 时的传输线参数。导体由铜制成（有关铜的 μ_c 和 σ_c，请参阅附录 B）。

答案：$R' = 2.07 \times 10^{-2}$ (Ω/m)，L′ = 0.14 (μ H/m)，G′ = 0，C′ = 80.3 (pF/m)。 （参见EM。）

2-3 传输线方程

传输线通常将一端的源连接到另一端的负载。然而，在考虑完整电路之前，我们将建立一般方程，将传输线上的电压和传输线承载的电流描述为时间 t 和空间位置 z 的函数。使用图 2-6(c) 的集总元件模型，我们首先考虑如图 2-8 所示的微分长度 Δz。量 υ(z,t) 和 i(z,t) 表示差分部分左端（节点 N）处的瞬时电压和电流，类似地，υ(z + Δz, t) 和 i(z + Δz,t) 表示节点处的相同量

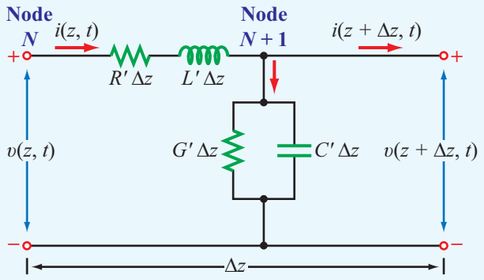


Figure 2-8 Equivalent circuit of a two-conductor transmission line of differential length Δz.

$(N+1)$, located at the right end of the section. Application of Kirchhoff's voltage law accounts for the voltage drop across the series resistance $R'\Delta z$ and inductance $L'\Delta z$:

$$v(z,t) - R'\Delta z\, i(z,t) - L'\,\Delta z\,\frac{\partial i(z,t)}{\partial t} - v(z+\Delta z,t) = 0. \quad (2.12)$$

Upon dividing all terms by Δz and rearranging them, we obtain

$$-\left[\frac{v(z+\Delta z,\,t) - v(z,t)}{\Delta z}\right] = R'\,i(z,t) + L'\,\frac{\partial i(z,t)}{\partial t}. \quad (2.13)$$

In the limit as $\Delta z \to 0$, Eq. (2.13) becomes a differential equation:

$$-\frac{\partial v(z,t)}{\partial z} = R'\,i(z,t) + L'\,\frac{\partial i(z,t)}{\partial t}. \quad (2.14)$$

Similarly, Kirchhoff's current law accounts for current drawn from the upper line at node $(N+1)$ by the parallel conductance $G'\,\Delta z$ and capacitance $C'\,\Delta z$:

$$i(z,t) - G'\,\Delta z\, v(z+\Delta z,t) - C'\,\Delta z\,\frac{\partial v(z+\Delta z,t)}{\partial t} - i(z+\Delta z,t) = 0. \quad (2.15)$$

Upon dividing all terms by Δz and taking the limit $\Delta z \to 0$, Eq. (2.15) becomes a second-order differential equation:

$$-\frac{\partial i(z,t)}{\partial z} = G'\,v(z,t) + C'\frac{\partial v(z,t)}{\partial t}. \quad (2.16)$$

The first-order differential equations (2.14) and (2.16) are the time-domain forms of the ***transmission-line equations***, known as the ***telegrapher's equations***.

Except for the last section of this chapter, our primary interest is in sinusoidal steady-state conditions. To that end, we make use of the phasor representation with a cosine reference, as outlined in Section 1-7. Thus, we define

$$v(z,t) = \mathfrak{Re}[\tilde{V}(z)\,e^{j\omega t}], \quad (2.17a)$$

$$i(z,t) = \mathfrak{Re}[\tilde{I}(z)\,e^{j\omega t}], \quad (2.17b)$$

where $\tilde{V}(z)$ and $\tilde{I}(z)$ are the phasor counterparts of $v(z,t)$ and $i(z,t)$, respectively, each of which may be real or complex. Upon substituting Eqs. (2.17a) and (2.17b) into Eqs. (2.14) and (2.16), and utilizing the property given by Eq. (1.62) that $\partial/\partial t$ in the time domain is equivalent to multiplication by $j\omega$ in the phasor domain, we obtain the following pair of equations:

$$-\frac{d\tilde{V}(z)}{dz} = (R' + j\omega L')\,\tilde{I}(z), \quad (2.18a)$$

$$-\frac{d\tilde{I}(z)}{dz} = (G' + j\omega C')\,\tilde{V}(z). \quad (2.18b)$$

(telegrapher's equations in phasor form)

2-4 Wave Propagation on a Transmission Line

The two first-order coupled equations (2.18a) and (2.18b) can be combined to give two second-order uncoupled wave equations: one for $\tilde{V}(z)$ and another for $\tilde{I}(z)$. The wave equation for $\tilde{V}(z)$ is derived by first differentiating both sides of Eq. (2.18a) with respect to z, resulting in

$$-\frac{d^2\tilde{V}(z)}{dz^2} = (R' + j\omega L')\frac{d\tilde{I}(z)}{dz}. \quad (2.19)$$

Then, upon substituting Eq. (2.18b) for $d\tilde{I}(z)/dz$, Eq. (2.19) becomes

$$\frac{d^2\tilde{V}(z)}{dz^2} - (R' + j\omega L')(G' + j\omega C')\,\tilde{V}(z) = 0, \quad (2.20)$$

or

$$\frac{d^2\tilde{V}(z)}{dz^2} - \gamma^2\,\tilde{V}(z) = 0, \quad (2.21)$$

(wave equation for $\tilde{V}(z)$)

where

$$\gamma = \sqrt{(R' + j\omega L')(G' + j\omega C')}. \quad (2.22)$$

(propagation constant)

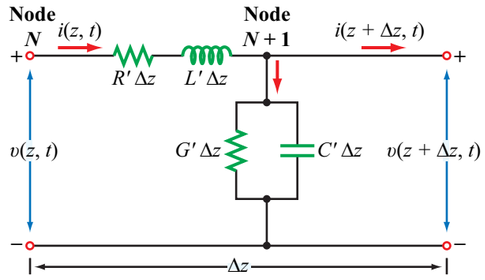


图 2-8 差分长度 Δz 的两导体传输线的等效电路。

(N + 1)，位于该部分的右端。应用基尔霍夫电压定律可计算串联电阻 R′Δz 和电感 L′Δz 两端的压降：

$$v(z,t) - R'\Delta z\, i(z,t) - L'\,\Delta z\,\frac{\partial i(z,t)}{\partial t} - v(z+\Delta z,t) = 0. \quad (2.12)$$

将所有项除以 Δz 并重新排列它们后，我们得到

$$-\left[\frac{v(z+\Delta z,\,t) - v(z,t)}{\Delta z}\right] = R'\,i(z,t) + L'\,\frac{\partial i(z,t)}{\partial t}. \quad (2.13)$$

在极限为 Δz → 0 时，式： (2.13) 变为微分方程：

$$-\frac{\partial v(z,t)}{\partial z} = R'\,i(z,t) + L'\,\frac{\partial i(z,t)}{\partial t}. \quad (2.14)$$

类似地，基尔霍夫电流定律解释了通过并联电导 G′ Δz 和电容 C ′ Δz 从节点 (N + 1) 处的上部线汲取的电流：

$$i(z,t) - G'\,\Delta z\, v(z+\Delta z,t) - C'\,\Delta z\,\frac{\partial v(z+\Delta z,t)}{\partial t} - i(z+\Delta z,t) = 0. \quad (2.15)$$

将所有项除以 Δz 并取极限 Δz → 0 时，方程： (2.15) 变为二阶微分方程：

$$-\frac{\partial i(z,t)}{\partial z} = G'\,v(z,t) + C'\frac{\partial v(z,t)}{\partial t}. \quad (2.16)$$

一阶微分方程（2.14）和（2.16）是传输线方程的时域形式，称为电报员方程。

除了本章的最后一节之外，我们的主要兴趣是正弦稳态条件。为此，我们使用带有余弦参考的相量表示，如第 1-7 节所述。因此，我们定义

$$v(z,t) = \mathfrak{Re}[\tilde{V}(z)\,e^{j\omega t}], \quad (2.17a)$$

$$i(z,t) = \mathfrak{Re}[\tilde{I}(z)\,e^{j\omega t}], \quad (2.17b)$$

其中，˜V (z) 和 ˜I(z) 分别是 v (z,t) 和 i(z,t) 的相量对应项，其中每个都可以是实数或复数。代入等式后。 (2.17a) 和 (2.17b) 代入等式。 (2.14) 和 (2.16)，并利用等式给出的属性。 (1.62) 时域中的∂ /∂ t等价于相量域中乘以jω，我们得到如下一对方程：

$$-\frac{d\tilde{V}(z)}{dz} = (R' + j\omega L')\,\tilde{I}(z), \quad (2.18a)$$

$$-\frac{d\tilde{I}(z)}{dz} = (G' + j\omega C')\,\tilde{V}(z). \quad (2.18b)$$

(相量形式的电报员方程)

2-4 传输线上的波传播

两个一阶耦合方程 (2.18a) 和 (2.18b) 可以组合起来得到两个二阶非耦合波动方程：一个用于 ˜V (z)，另一个用于 ˜I(z)。 ˜V (z) 的波动方程是通过首先对方程两边求导得出的。 (2.18a) 关于 z，导致

$$-\frac{d^2\tilde{V}(z)}{dz^2} = (R' + j\omega L')\frac{d\tilde{I}(z)}{dz}. \quad (2.19)$$

然后，将方程代入后。 (2.18b) 对于 d ˜I(z)/dz，方程(2.19) 变为

$$\frac{d^2\tilde{V}(z)}{dz^2} - (R' + j\omega L')(G' + j\omega C')\,\tilde{V}(z) = 0, \quad (2.20)$$

或者

$$\frac{d^2\tilde{V}(z)}{dz^2} - \gamma^2\,\tilde{V}(z) = 0, \quad (2.21)$$

(wave equation for $\tilde{V}(z)$)

在哪里

$$\gamma = \sqrt{(R' + j\omega L')(G' + j\omega C')}. \quad (2.22)$$

(propagation constant)

Application of the same steps to Eqs. (2.18a) and (2.18b) in reverse order leads to

$$\frac{d^2\tilde{I}(z)}{dz^2} - \gamma^2\tilde{I}(z) = 0. \qquad (2.23)$$

(wave equation for $\tilde{I}(z)$)

The second-order differential equations (2.21) and (2.23) are called ***wave equations*** for $\tilde{V}(z)$ and $\tilde{I}(z)$, respectively, and γ is called the ***complex propagation constant*** of the transmission line. As such, γ consists of a real part α, called the ***attenuation constant*** of the line with units of Np/m, and an imaginary part β, called the ***phase constant*** of the line with units of rad/m. Thus,

$$\gamma = \alpha + j\beta \qquad (2.24)$$

with

$$\alpha = \Re\mathrm{e}(\gamma) = \Re\mathrm{e}\left(\sqrt{(R'+j\omega L')(G'+j\omega C')}\right) \quad \text{(Np/m)}, \qquad (2.25a)$$

(attenuation constant)

$$\beta = \Im\mathrm{m}(\gamma) = \Im\mathrm{m}\left(\sqrt{(R'+j\omega L')(G'+j\omega C')}\right) \quad \text{(rad/m)}. \qquad (2.25b)$$

(phase constant)

2-4.1 Phasor-Domain Solution

In Eqs. (2.25a) and (2.25b), we choose the square-root solutions that give positive values for α and β. For passive transmission lines, α is either zero or positive. Most transmission lines, and all those considered in this chapter, are of the passive type. The gain region of a laser is an example of an active transmission line with a negative α.

The wave equations (2.21) and (2.23) have traveling wave solutions of the form:

$$\tilde{V}(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z} \quad \text{(V)}, \qquad (2.26a)$$

$$\tilde{I}(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z} \quad \text{(A)}. \qquad (2.26b)$$

As shown later, the $e^{-\gamma z}$ term represents a wave propagating in the $+z$ direction, while the $e^{\gamma z}$ term represents a wave propagating in the $-z$ direction (**Fig. 2-9**). Verification that these are indeed valid solutions is easily accomplished by substituting the expressions given by Eqs. (2.26a) and (2.26b), as well as their second derivatives, into Eqs. (2.21) and (2.23).

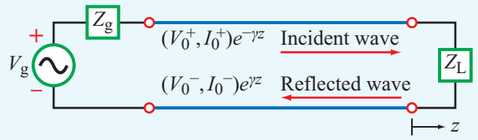


Figure 2-9 In general, a transmission line can support two traveling waves: an incident wave (with voltage and current amplitudes (V_0^+, I_0^+)) traveling along the $+z$ direction (towards the load) and a reflected wave (with (V_0^-, I_0^-)) traveling along the $-z$ direction (towards the source).

In their present form, the solutions given by Eqs. (2.26a) and (2.26b) contain four unknowns: the wave amplitudes (V_0^+, I_0^+) of the $+z$ propagating wave and (V_0^-, I_0^-) of the $-z$ propagating wave. We can easily relate the current wave amplitudes, I_0^+ and I_0^-, to the voltage wave amplitudes, V_0^+ and V_0^-, by using Eq. (2.26a) in Eq. (2.18a) and then solving for the current $\tilde{I}(z)$. The process leads to

$$\tilde{I}(z) = \frac{\gamma}{R'+j\omega L'}[V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}]. \qquad (2.27)$$

Comparison of each term with the corresponding term in Eq. (2.26b) leads us to conclude that

$$\frac{V_0^+}{I_0^+} = Z_0 = \frac{-V_0^-}{I_0^-}, \qquad (2.28)$$

where

$$Z_0 = \frac{R'+j\omega L'}{\gamma} = \sqrt{\frac{R'+j\omega L'}{G'+j\omega C'}} \quad (\Omega), \qquad (2.29)$$

is called the ***characteristic impedance*** of the line.

► It should be noted that Z_0 is equal to the ratio of the voltage amplitude to the current amplitude for each of the traveling waves individually (with an additional minus sign in the case of the $-z$ propagating wave), but it is not equal to the ratio of the total voltage $\tilde{V}(z)$ to the total current $\tilde{I}(z)$ unless one of the two waves is absent. ◄

It seems reasonable that the voltage-to-current ratios of the two waves V_0^+/I_0^+ and V_0^-/I_0^- are both related to the same quantity, namely Z_0, but it is not immediately obvious as

对等式应用相同的步骤。 (2.18a) 和 (2.18b) 按相反顺序可得

$$\frac{d^2\tilde{I}(z)}{dz^2} - \gamma^2\tilde{I}(z) = 0. \qquad (2.23)$$

(wave equation for $\tilde{I}(z)$)

二阶微分方程（2.21）和（2.23）分别称为˜V（z）和Ĩ（z）的波动方程，γ称为传输线的复传播常数。因此，γ由实部α（称为线的衰减常数，单位为Np/m）和虚部β（称为线的相位常数，单位为rad/m）组成。因此，

$$\gamma = \alpha + j\beta \qquad (2.24)$$

和

$$\alpha = \Re\mathrm{e}(\gamma) = \Re\mathrm{e}\left(\sqrt{(R'+j\omega L')(G'+j\omega C')}\right) \quad \text{(Np/m)}, \qquad (2.25a)$$

(attenuation constant)

$$\beta = \Im\mathrm{m}(\gamma) = \Im\mathrm{m}\left(\sqrt{(R'+j\omega L')(G'+j\omega C')}\right) \quad \text{(rad/m)}. \qquad (2.25b)$$

(phase constant)

2-4.1 相量域解

在方程式中。 (2.25a) 和 (2.25b) 中，我们选择为 α 和 β 提供正值的平方根解。对于无源传输线，α 为零或正。大多数传输线以及本章中考虑的所有传输线都是无源类型。激光器的增益区域是具有负 α 的有源传输线的示例。

波动方程 (2.21) 和 (2.23) 具有以下形式的行波解：

$$\tilde{V}(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z} \quad \text{(V)}, \qquad (2.26a)$$

$$\tilde{I}(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z} \quad \text{(A)}. \qquad (2.26b)$$

如后所示，$e^{-\gamma z}$项表示沿+z方向传播的波，而$e^{\gamma z}$项表示沿−z方向传播的波（图2-9）。通过替换方程式给出的表达式，可以轻松验证这些确实是有效的解决方案。将（2.26a）和（2.26b）及其二阶导数代入方程。 （2.21）和（2.23）。

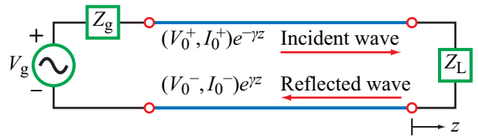


图 2-9 一般来说，传输线可以支持两种行波：沿 +z 方向（朝向负载）传播的入射波（电压和电流幅度为 (V_0^+, I_0^+)）和沿 −z 方向（朝向源）传播的反射波（具有 (V_0^-, I_0^-)）。

以目前的形式，方程给出的解决方案。 (2.26a)和(2.26b)包含四个未知数：+z传播波的波幅(V_0^+，I_0^+)和−z传播波的(V_0^-，I_0^-)。通过使用等式，我们可以轻松地将电流波幅度 I_0^+ 和 I_0^- 与电压波幅度 V_0^+ 和 V_0^- 联系起来。方程（2.26a） (2.18a) 然后求解电流的 Ĩ(z)。该过程导致

$$\tilde{I}(z) = \frac{\gamma}{R'+j\omega L'}[V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}]. \qquad (2.27)$$

每一项与等式中相应项的比较。 (2.26b) 使我们得出结论

$$\frac{V_0^+}{I_0^+} = Z_0 = \frac{-V_0^-}{I_0^-}, \qquad (2.28)$$

在哪里

$$Z_0 = \frac{R'+j\omega L'}{\gamma} = \sqrt{\frac{R'+j\omega L'}{G'+j\omega C'}} \quad (\Omega), \qquad (2.29)$$

称为线路的特性阻抗。

► 应该注意的是，Z_0 分别等于每个行波的电压幅度与电流幅度之比（在 −z 传播波的情况下带有附加负号），但它不等于总电压 ˜V (z) 与总电流 Ĩ(z) 的比率，除非两个波之一不存在。 ◄

两个波 V_0^+/I_0^+ 和 V_0^-/I_0^- 的电压电流比都与相同的量（即 Z_0）相关，这似乎是合理的，但并不立即显而易见，因为